

Use of dual-frequency microwave links for measuring path-averaged rainfall

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Received 21 November 2002; accepted 24 March 2003; published 9 August 2003.

[1] Results are presented from an experiment designed to test the use of dual-frequency microwave attenuation for measuring path-averaged rainfall. In general, the relation between attenuation and path-averaged rainfall is nonlinear. However, for carefully chosen pairs of frequencies, the relation between the difference in attenuations and path-averaged rainfall is well represented by a straight line through the origin. The slope of this line is essentially unaffected by variations in the (unknown) drop size distribution, drop shape, and drop temperature. This paper describes the results obtained from a pair of links operating at 12.8 GHz and 17.6 GHz on a 23.3 km path. Results are shown for three individual events, including comparisons with rainfall estimates obtained from more traditional sources: a local rain gauge network and an adjacent weather radar. We report summary results for 112 events recorded on this path during almost 2 years of observation, and also for 52 events recorded on a nearby 13.9 km path operating at 13.9 GHz and 22.9 GHz. In both cases the results suggest that dual-frequency links could serve as useful additional tools for the measurement of rainfall, particularly in locations such as urban areas and obscured mountain valleys, where traditional measuring instruments struggle to provide accurate estimates. *INDEX TERMS:* 1854 Hydrology: Precipitation (3354); 1794 History of Geophysics: Instruments and techniques; 3360 Meteorology and Atmospheric Dynamics: Remote sensing; 6964 Radio Science: Radio wave propagation; *KEYWORDS:* microwave links, attenuation difference, rainfall estimation, baseline calibration

Citation: Rahimi, A. R., A. R. Holt, G. J. G. Upton, and R. J. Cummings, Use of dual-frequency microwave links for measuring path-averaged rainfall, *J. Geophys. Res.*, 108(D15), 4467, doi:10.1029/2002JD003202, 2003.

1. Introduction

[2] Real-time measurement of rainfall over a catchment has usually been attempted by a network of telemetered rain gauges [O'Connell *et al.*, 1979], or, more recently, by ground-based radar [Collier, 1989]. This paper is concerned with a new method of estimating rainfall, the dual-frequency microwave link, which provides continuous-time measurements of path-averaged rainfall close to ground level. Links can provide a useful complement to traditional methods and will be of particular value in situations where rain gauges and radar have difficulties. A brief review of the difficulties faced by these traditional methods is given in section 2.

[3] This paper introduces the theory underlying the new method, including the choice of frequencies (section 3). We present details of an experiment conducted in the Bolton region of north-west England to test the method (section 4). We describe the practical implementation of the method (section 5) and summarize the results of nearly 2 years of observation (sections 6 and 7). We discuss the issues of link length and coverage (section 8). We conclude (section 9) that the link provides reliable estimates of path-averaged

rainfall, with storm estimates generally lying within 15% of comparable estimates. It appears that a single link could give a useful estimate of total rainfall for a catchment area as large as 100 km².

2. Difficulties With Traditional Methods

[4] Rain gauge networks are expensive to set up and telemeter, and the gauges require regular maintenance visits to keep them clear from obstruction [e.g., Wood *et al.*, 2000]. Tipping-bucket gauges are by far the most common type of gauge [Sevruk, 2002]. As with all gauges, they are not free from bias problems related both to the intensity of the rainfall and the wind field around the gauge, and are only accurate to about 10% [e.g., Sevruk, 1996]. In urban areas and mountain valleys, in particular, gauges can be very difficult to site since they should be neither sheltered nor exposed. Gauges are also liable to vandalism and interference by animals. Moreover, a rain gauge makes what is virtually a point measurement. Since rain fields can have strong gradients, particularly in convective storms [e.g., Semplak, 1971], a series of point measurements may give a poor reflection of the areal rainfall in the catchment.

[5] Radar too, has its problems [Collier, 1989]. A radar does not measure rainfall, but reflected energy, and there is no unique relationship between these two quantities. Since a

radar normally scans at an elevation angle of at least 0.5° , it is scanning above the surface of the Earth with the volume scanned increasing with the square of its distance from the radar. Such scanning may be subject to obscuration due to the local topography, and the data collected will not be confined to reflections from rain. There may also be “clutter”, energy reflected from the ground or insects, etc., and reflections from other forms of precipitation such as hail or melting particles. Nevertheless, radar can provide excellent spatial resolution over a wide area, typically a circle of at least 100 km radius.

3. Attenuation and Rainfall

[6] When an electromagnetic wave of wavelength λ and wave number $k_0 = \frac{2\pi}{\lambda}$, impinges on a region of hydrometeors, the transmitted wave will be attenuated, since energy will be scattered and absorbed by the hydrometeors [Van de Hulst, 1981]. We assume that the number of drops with equivolume diameter in the range $(D, D + dD)$ is $N(D)dD$, and that the (complex) forward scattering amplitude for scattering by a drop of diameter D is $f(D)$. Then, with

$$K = k_0 + \frac{2\pi}{k_0} \int_0^{D_{\max}} f(D)N(D)dD, \quad (1)$$

the attenuation due to rain, in dB per kilometer, is given by

$$A = 8.686 \times 10^5 \text{Im}(K) \quad (2)$$

where D and f are in cm, K and k_0 are in cm^{-1} and $N(D)$ is in cm^{-4} .

[7] The relationship between rainfall and attenuation has been considered by many authors. For example, *Atlas and Ulbrich* [1977] noted a near-linear relation between rainfall and attenuation at 35 GHz, while *Jameson* [1991] shows that the relation at 25 GHz is even closer (with rain rate proportional to attenuation raised to the power 0.99) and is unaffected by variations in the drop size distribution. These precise theoretical relations become less useful when experimental error is taken into account. Jameson considered the measurement errors to be expected on a 1 km link and concluded that the attenuation at these frequencies would only be a useful guide for path-averaged rain rates in excess of 1 mm/hr, though the sensitivity would be enhanced by using a longer link. Jameson also noted that the attenuation differences between the three frequencies 9 GHz, 25 GHz and 38 GHz, were capable of providing useful estimates. This is confirmed by *Rincon and Lang* [2002] who report results for a dual-frequency microwave link experiment using 25 GHz and 38 GHz with a 2.3 km link. They studied 15 rain events amounting to around 30 hours of rainfall, and suggest that, given certain assumptions, it would be feasible to estimate the dsd, and hence the rainfall along the link path. That an attenuation difference could be useful for measuring rainfall had also been noted during an experiment monitoring the 12.5 GHz and 19.8 GHz data from the Olympus Satellite [Hardaker et al., 1997], and it was that realization that led to the experiment described in this paper.

[8] These studies of the possibilities inherent in the use of attenuation difference raise the question as to which pairs of frequencies would be most suitable for the purpose of measuring path-averaged rainfall. An ideal remote sensing system would give results which are independent of unknowns such as the dsd, $N(D)$, the temperature, the shape of the hydrometeors and their orientation in space.

[9] To judge whether a result is insensitive to these unknowns, one can vary the assumptions about the underlying parameters, and investigate the sensitivity of the results to these assumptions. For our investigations we have assumed a gamma dsd of the form

$$N(D) = N_0 D^\mu \exp\left(-\left(3.67 + \mu\right) \frac{D}{D_0}\right), \quad (3)$$

with the values of N_0 and D_0 being either those given by *Ulbrich* [1983] or those given by *Testud et al.* [2001]. For each case we have considered seven values of μ ($-1, 0, \dots, 5$), three temperatures ($5^\circ\text{C}, 10^\circ\text{C}, 15^\circ\text{C}$), and three raindrop shapes: those of *Goddard and Cherry* [1984], *Beard and Chuang* [1987], and *Morrison and Cross* [1974]. We considered that these three shapes adequately spanned the likely range of drop shapes that might be encountered. We assumed $D_{\max}$ equals 0.7 cm and that the drops were oriented with their symmetry axes vertical.

[10] A detailed description of the resulting calculations across the frequency range has been given by *Holt et al.* [2003]. For each form of the gamma distribution, the set of 63 curves was obtained for attenuation difference as a function of rain rate (the 63 curves are those resulting from the different combinations of the seven gamma distributions, three temperatures and three drop shapes). For each ensemble of curves, the gradient and intercept of the least squares linear fit was calculated.

[11] As an example of the results obtained, we consider the case of 12.8 GHz and 17.6 GHz with vertical polarization (a preference expressed by the system installers). The resulting 63 curves for the Ulbrich form of the dsd are illustrated in Figure 1.

[12] For these Ulbrich curves it was found that a difference in attenuation of 1 dB/km corresponds to a rain rate of between 26 mm/hr and 36 mm/hr. The unconstrained least squares fit gives a gradient of 0.0323 (dB/km)/(mm/hr) and an intercept of -0.0093 dB/km. The least squares fit constrained to pass through the origin gives a gradient of 0.0320 (dB/km)/(mm/hr). The corresponding values of the gradient and intercept for the Testud case are 0.0329 (dB/km)/(mm/hr) and -0.0097 dB/km, with the gradient for the constrained fit being 0.0325 (dB/km)/(mm/hr). The assumption of a single line passing through the origin is clearly reasonable; in this paper we have assumed a constrained fit with a gradient of 0.032 (dB/km)/(mm/hr). Since we have included extremes of drop shape and drop size distribution, the range of variation displayed suggests that we can hope to estimate the path-averaged rain rate to within about 15% of the true value.

[13] It should be noted here that precipitation along a path may not be rain alone, but could be hail, or, if the temperature is low enough, sleet or snow. Attenuation in snow is negligible. Attenuation in sleet can be substantially greater than that in rain, and the method then over-estimates

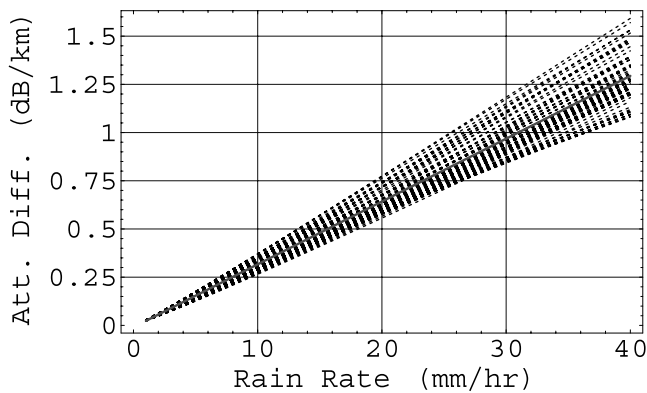


Figure 1. Dependence of attenuation difference on rain rate for the case of 12.8 GHz and 17.6 GHz, each with vertical polarization. The graph shows 63 separate curves corresponding to different combinations of drop temperature, drop shape and the μ parameter of the Ulbrich form of the gamma dsd.

the amount of precipitation (though our recent research suggests that we can detect when this occurs in the link data).

4. Bolton Experiment

4.1. Overview

[14] In order to test the applicability of using dual-frequency microwave links to monitor rainfall and provide input to a hydrological model of an urban catchment [Robbins and Collier, 2002], a collaborative experiment was established in north-west England, centered on the town of Bolton. Purpose-built dual-frequency links were installed by the Rutherford Appleton Laboratory. A network of up to 22 tipping-bucket rain gauges provided conventional rainfall estimates, and two automatic weather stations recording standard climatological variables such as temperature, pressure and wind speed were installed by the UK Met Office who also made available selected radar data from the nearby Hameldon Hill Weather Radar.

[15] The locations of the equipment are shown in Figure 2. The area contains both a river valley (covered by the long link to the right of the figure) and an urban area (covered by the remaining links).

4.2. Microwave Links

[16] The microwave links were custom-built at the Rutherford Appleton Laboratory to achieve the level of stability and reliability required for the experiment. Attenuations can be measured to a precision of approximately ± 0.5 dB with a 40 dB dynamic range. A worst-case scenario would therefore result in a potential error of ± 1 dB for the attenuation difference. However, other sources of signal fluctuation such as atmospheric and multipath effects will also be present in the data. We describe how we correct for these effects in section 5. For further technical details regarding the equipment and selection of frequencies, see Holt et al. [2000].

[17] For the most part we shall be reporting results from the river-valley link, which provided dual-frequency data (at 12.8 GHz and 17.6 GHz) between February 2000 and

October 2001. The link runs for 23.3 km almost due South along the valley of the river Irwell, with the transmitter site at a height of 396 m (on Hameldon Hill) and the receiver site at 130 m (on Clarke's Hill). As required, there is a clear line-of-sight between transmitter and receiver. However, the beam is about 30 m from the ground at the closest point and the proximity of the ground doubtless leads to multipath effects. Reflections from the sides of the narrow valley may also occur.

[18] The link data were sampled every second, although for rainfall estimation purposes one-minute averages are used. Link traces for a typical rainy day (17 May 2001), are presented in Figure 3. Attenuations at the two single frequencies are plotted (top) along with the corresponding differences (bottom). For the higher frequency (17.6 GHz) the peak attenuation during the event is in excess of 20 dB. The corresponding peak attenuation difference is about 8 dB. When the weather is dry (e.g., toward the end of the day), both frequencies experience very little attenuation.

4.3. Rain Gauges

[19] The experiment used information from 22 gauges, of which eleven were supplied by the UK Environment Agency, nine were supplied by United Utilities and two by the UK Met Office. The rain gauges used in the experiment were all 0.2 mm tipping-bucket gauges, recording tip-times to the nearest second. To facilitate comparison with the one-minute attenuation averages, the tip-time data were converted to one-minute rain rates, assuming a uniform rain rate between recorded tips.

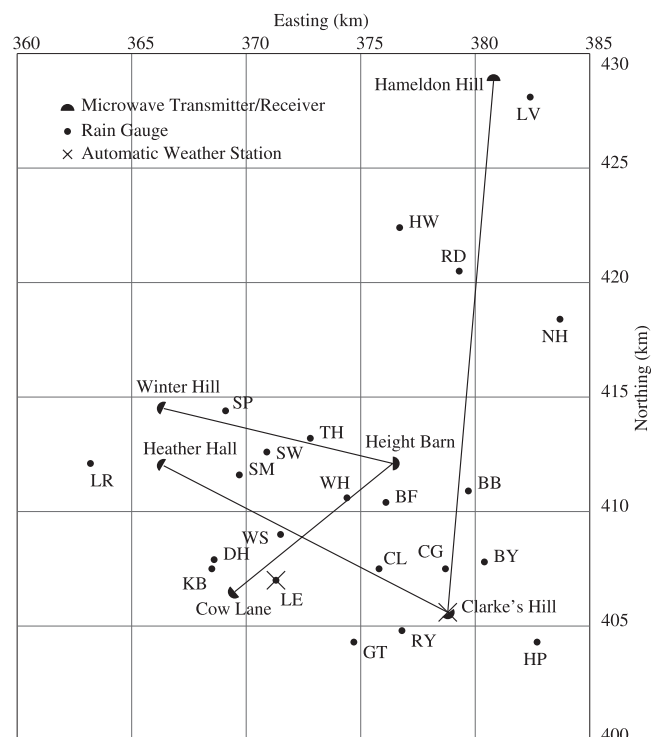


Figure 2. Map of experiment area. The co-ordinates are those of the UK Ordnance Survey. The positions are indicated for the rain gauges, the two automatic weather stations and the four experimental links.

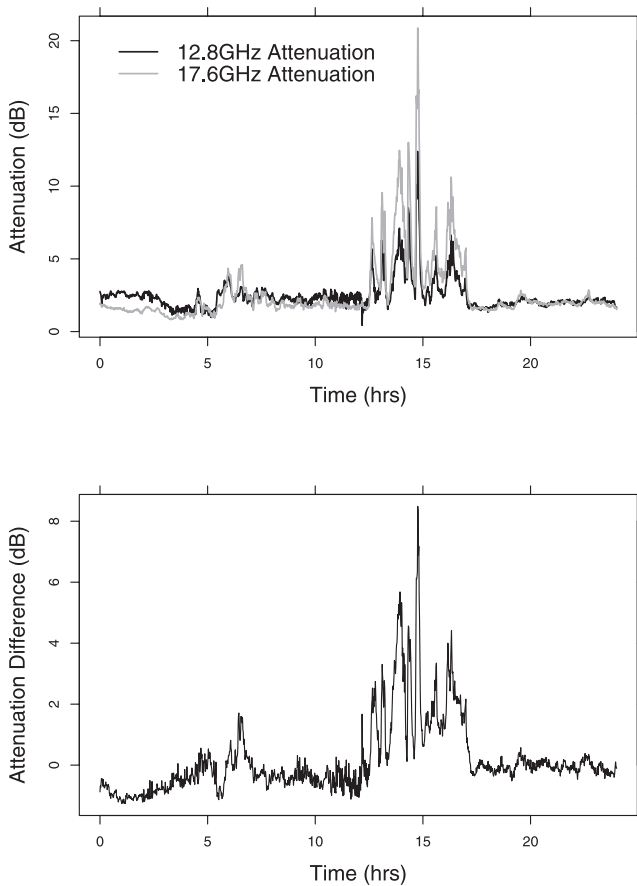


Figure 3. Microwave link attenuations for 17 May 2001: Single frequency attenuations (top) and their difference (bottom).

[20] One motivation for investigating an alternative method of measuring rainfall is unreliability in current techniques. Tipping-bucket rain gauges are vulnerable to a host of potential problems. In the course of the Bolton experiment, we have observed inaccurate rain gauge data as a consequence of blockages (many sources, including grass cuttings, soil and dirt, hail), mechanical faults (e.g., with the tipping mechanism), frozen gauges, problems with data loggers and computer downloads, and inappropriate siting. At one point in the experiment (December 2000), more than one third of the rain gauges were producing unusable data.

[21] We can see some typical examples of faulty gauges in Figure 4 which illustrates the one-minute tip data for the gauges (rows) on 13 December 2000. Each vertical line indicates that a tip was recorded in the corresponding minute. Gauges LE, WH, BF and CG were blocked (but in different ways). Gauge CH had its top blown off in high winds and was recording up to 12 tips per minute in the early morning. Gauges LV and HW suffered computer and data logger problems while gauge KB was inappropriately sited.

[22] It is clearly important to use only good-quality rain gauge data in any analysis or comparisons with link data. As a consequence of our analyses of the Bolton rain gauge data we have devised a variety of diagnostic procedures

designed to automatically identify faulty gauges. These procedures are described in a forthcoming paper [Upton and Rahimi, 2003].

4.4. Radar

[23] A small amount of data from the Met Office C-Band Weather Radar situated at Hameldon Hill was made available for the experiment. Since Hameldon Hill is the location of the transmitter for one of the microwave links (see Figure 2), the radar looks along roughly the same path as that link. The radar data made available were in Cartesian form (2 km-squares), with the radar scanning every five minutes.

13 / Dec / 2000 (Tips of the Day)

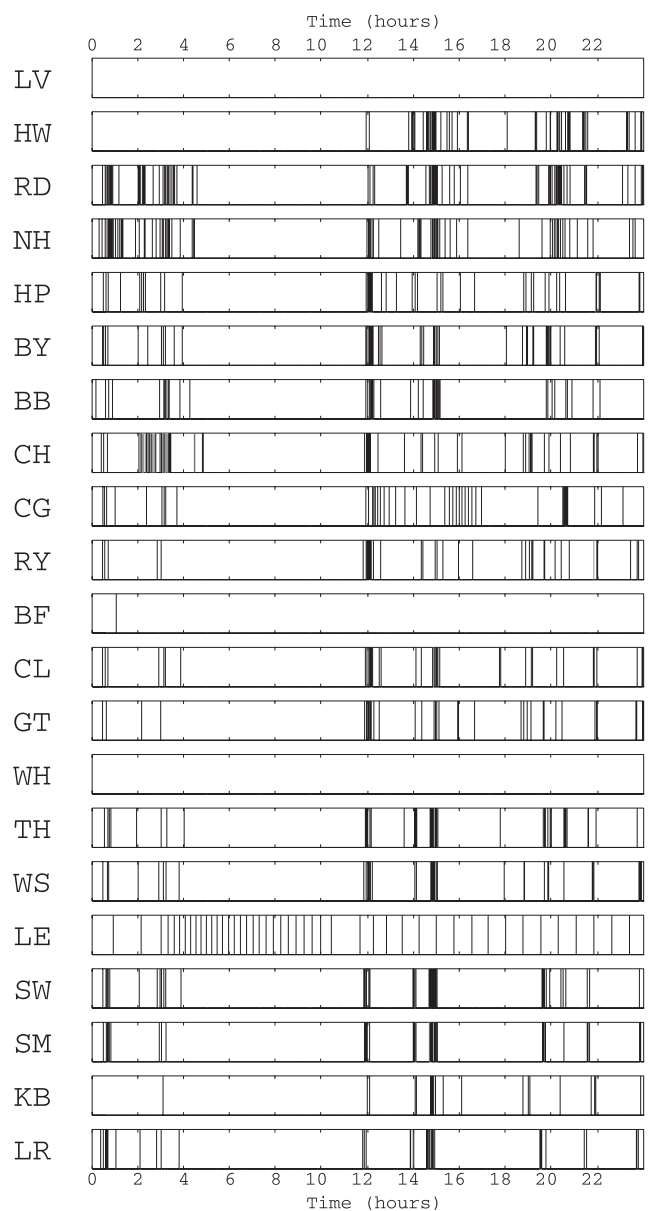


Figure 4. Rain gauge tips for 13 December 2000: Each row corresponds to a gauge; each vertical line corresponds to a minute in which a tip was recorded.

5. Rain Rate Estimation

5.1. Link Rain Rate Estimation

[24] Path-averaged rainfall is estimated from the attenuation difference using the linear relationship discussed in section 3. However, the link data require calibration before this estimation can be carried out. Possible calibration procedures are described in the next section. The merits of these procedures can be judged by comparing the resulting rainfall estimates with those obtained from rain gauges and from radar data.

5.2. Link Calibration Using Atmospheric Variables

[25] In addition to attenuation caused by raindrops, microwaves are attenuated by gases occurring naturally in the air, particularly water vapor [Brussaard and Watson, 1995]. For a terrestrial link this attenuation depends on its frequency and path length, as well as the prevailing atmospheric conditions. The *International Telecommunication Union (ITU)* [1997] has produced a recommendation for the estimation of attenuation by atmospheric gases for terrestrial microwave links. This is a frequency-dependent adjustment that involves temperature, humidity and pressure.

[26] Multipath effects will also contribute to fluctuating attenuation levels. In order to measure rainfall accurately using the attenuation difference method it is necessary to consider only the attenuation caused by the rain itself. We therefore require methods to correct or calibrate link data to take this into account.

[27] For the 17.6 GHz link, analysis of periods of dry weather (hours in which no rain gauge recorded rain) has revealed that the ITU adjustment for gaseous absorption does not account for all the attenuation experienced. Figure 5 (top) shows the dry-period attenuations for the 17.6 GHz link from February 2000 until the end of July 2001 before correction for gaseous absorption. Figure 5 (middle) shows the same data after the ITU correction has been applied using the relevant climate data from the weather stations. The evident annual cycle that remains may be a consequence of terrain-induced multipath effects. Some of the effects can also be attributed to the climate data not being appropriate for the entire link path. The northern end of the 23.3 km link is over 250 meters higher than the southern end (where the weather stations are located).

[28] Multiple-regression analysis was conducted on 1 year of ITU-corrected dry-period attenuations for each frequency in an attempt to model the observed residual attenuation. Using meteorological variables recorded by the weather stations the analysis resulted in a model that accounts for much of the attenuation not induced by rain. This is illustrated in Figure 5 (bottom) for the 17.6 GHz frequency which shows dry-period attenuations after correction using the regression model. The model, which involves atmospheric variables along with a day/night factor, explains about 70% of the dry-period variability. In terms of magnitude relative to the raw data, corrections at the higher frequency range from about 1 dB in the winter months to 2.5 dB in the middle of summer. For the lower frequency the corrections are generally between 0.5 dB and 1.5 dB.

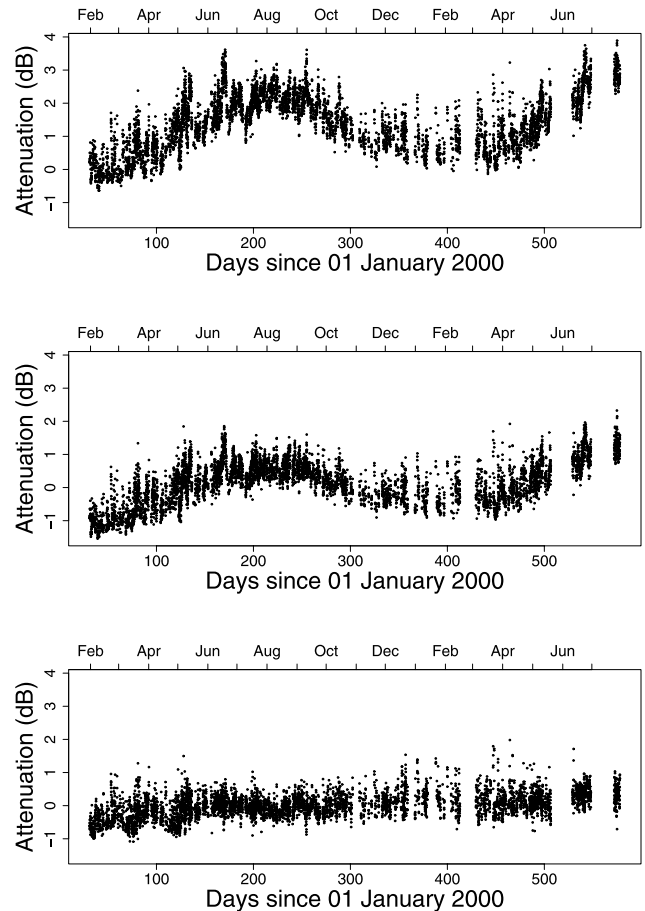


Figure 5. Dry period attenuations (17.6 GHz): Raw (top); ITU-corrected (middle); regression-corrected (bottom).

[29] However, while the regression method is successful at correcting for long-term variability in attenuation, it does not account well for shorter-term fluctuations, and we have therefore adopted an alternative approach which also has the advantage of not requiring knowledge of atmospheric or other variables.

5.3. Link Calibration Using Local Corrections

[30] Rainy periods are typically quite short. A reasonable assumption therefore is that the atmospheric and other factors that were influencing the attenuation in the preceding dry period will change little in the subsequent wet period, though it will, of course, have started to rain.

[31] We wish to adjust the baseline attenuation level by reference to the attenuation experienced in dry periods. We therefore require an operationally viable method for identifying these periods. We discuss two such methods in the next sections. Using either method, we propose a rolling calibration system in which the mean baseline attenuation level for each frequency is constantly updated during spells of dry weather. The most recent dry-weather baseline levels would be used to determine the attenuations in the subsequent wet period. As a refinement, the estimates of the attenuations during the rainy period can later be revised by reference to the baseline in the subsequent dry period.

[32] In these local-correction schemes it is essential that a period that has been identified as being dry is indeed dry. The consequence of misclassifying a wet period as being dry would be that a wet-period attenuation would be subtracted from all the subsequent rain attenuations. This would result in considerable underestimation of the event rainfall. By contrast, misclassifying a dry period as being wet is less serious, since the amount of rain that is estimated for that period is sure to be small (since the attenuation difference itself will be small as it is not, in fact, raining). The methods for dry period identification therefore concentrate on avoiding the misclassification of a wet period as being dry.

5.3.1. Identification of Wet and Dry Periods Using Correlation

[33] The correlation between attenuations at two frequencies tends to be high in rain because rain falling on the link will cause well-defined changes in attenuation in similar fashions for both frequencies. The apparent correlation will depend on the numbers of observation pairs and the time intervals to which they refer. We have used sequences of 15 pairs, with the individual values being 60-second mean attenuations. We define dry periods as being 15-minute periods in which no gauge recorded a tip, and wet periods as being 15-minute periods in which at least two gauges recorded a tip (this guards against faulty gauges).

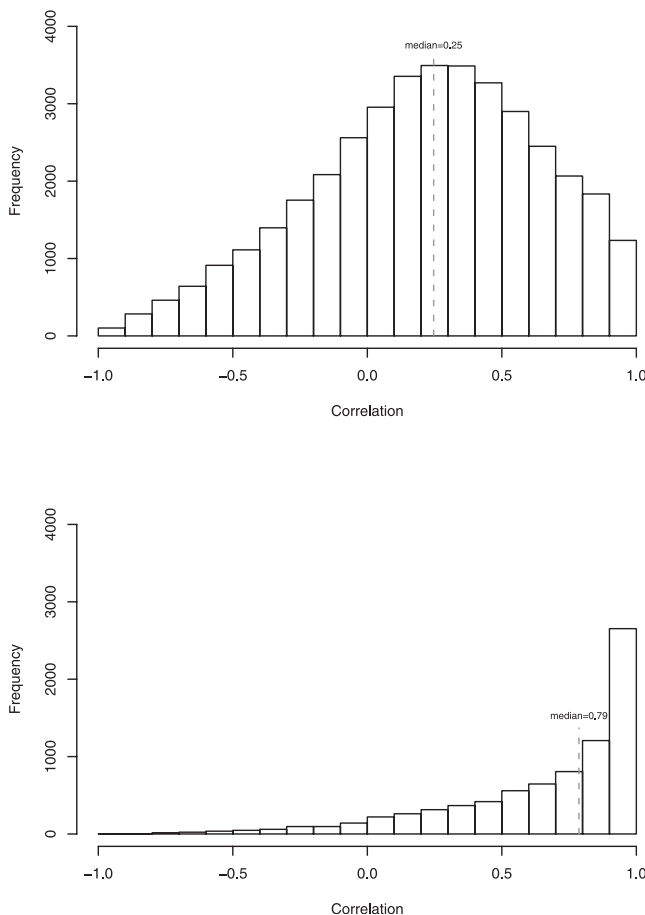


Figure 6. Distribution of attenuation correlation (15-minute periods): Dry periods (top) and wet periods (bottom).

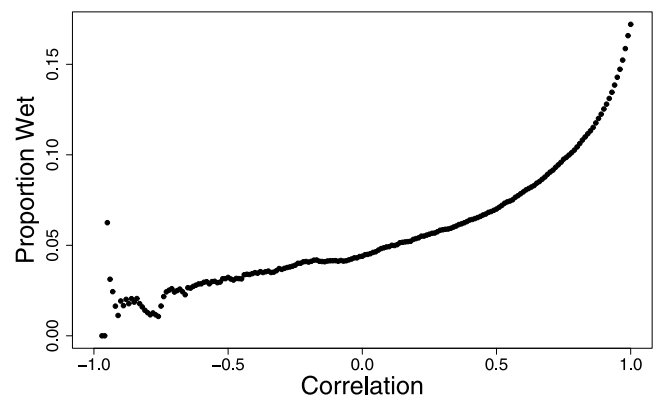


Figure 7. Probability of wetness: Proportion of 15-minute periods identified as dry when actually wet, for given correlation thresholds.

[34] As expected, Figure 6 shows that there is a marked difference in the two correlation distributions. The distribution of the correlation in wet periods is highly skewed toward 1, with a median value of 0.79. This is in contrast to the dry-period correlations, which are more symmetrically distributed, with a median of 0.25.

[35] The considerable difference between the two medians suggests that we might be reasonably accurate in classifying 15-minute periods as wet or dry using only the observed value of the correlation (though other information would obviously help). To decide on an appropriate value r_c , the critical correlation value above which a period is classified as wet, a combination of two methods has been used. Figure 7 plots the proportion of periods that would be incorrectly identified as being dry as r_c is varied. Although we wish this proportion to be small, choosing too small a value for r_c would result in infrequent updating of the baseline levels and a consequent reduction in their relevance. For example, from Figure 7 we see that if we set r_c to be -0.50 , the probability of misclassifying a wet period would be less than 0.03. However, from Figure 6 we find that this threshold level would result in only 6% of dry periods actually being identified as dry. We therefore need to strike a balance.

[36] In order to set the critical level objectively, rainfall estimation was carried out for various values of r_c . For each value, the root mean square error (RMSE) was calculated, using a comparison of the 15-minute accumulations estimated from the rain gauge data with those estimated by the link. The threshold value, r_c , that minimized the RMSE (using eighteen months of data) was 0.60. From Figures 6 and 7 we can see that this choice results in approximately 80% of dry periods being used for calibration, with just over 7.5% of wet periods being misclassified as dry.

5.3.2. Operational Details

[37] In operation the mean attenuation levels at the two frequencies (low and high), and their correlation, r , are calculated every 15 min. If $r \geq r_c$, then the period is judged to be wet. However, if $r < r_c$, then a cautious approach is taken and this period is initially described as provisionally dry; it becomes a confirmed dry period only if $r < r_c$ in the next 15-minute period. If that is the case, then the mean attenuation levels are taken as being the current baseline levels. In the same way, we classify the “dry” period that

occurs immediately after a wet period as being conditionally dry. These rules effectively provide a 15-minute buffer between wet periods and dry periods.

[38] A variety of different durations were considered for the dry-period identification process, including 5, 20, 30 and 60 minutes. The final choice of 15 min was made on the basis that it was sufficiently short to serve as an indicator of “local” attenuation levels while being long enough to provide a reasonably reliable estimate of the correlation.

5.3.3. Use of Rain Detectors

[39] The method described in the previous sections relies entirely on correlation, and hence entirely on links, to distinguish between wet and dry periods. A more reliable method incorporates additional information from some form of rain detector.

[40] For future implementation of the procedure we suggest that simple rain detectors be included with the equipment at each end of the link. If either detector reports that it is raining then that 15-minute period is a wet period. As in the previous method, we also describe a period as being wet if the correlation between the two attenuations exceeds some critical value, r_c . In this case a high value of r_c is appropriate (we used 0.8) since we require strong evidence of mid-link rainfall if rain is not being detected at either end of the link. The likelihood of this occurring will increase with link length and in convective rainfall.

[41] For the Bolton experiment, for which rain detectors were not installed, the information from the nearest rain gauge to each end of the link was used as a surrogate. The principal difficulty with using tipping-bucket rain gauges to detect rain is that we have no idea when the rain that causes the bucket to tip, actually falls. Taking a cautious approach, we have assumed that a tip implies that there may have been rain at any time in the preceding 30 min, and so the previous two 15-minute periods are taken to be wet, as are any 15-minute periods in which a tip is actually recorded. All other periods are taken to be dry unless the correlation between attenuations is above the threshold level, or (as insurance) the previous period was wet.

5.4. Path-Averaging of Rain Gauge Data

[42] Rainfall estimates derived from a microwave link are path-averaged, whereas rain gauges provide point estimates. To allow comparisons, the rain gauge data must be converted into path-averaged form. We used a standard Thiessen approach [Upton and Fingleton, 1985, chap. 1] allocating each portion of the link to its nearest gauge. The gauge estimates reported here are therefore weighted averages of the records from the nearest working gauges. For the Hameldon Hill to Clarkes Hill link these were usually the gauges identified as CG, BB, RD and LV on Figure 2.

[43] For events such as summer thunderstorms in which rainfall is spatially nonuniform, such interpolation methods may not result in accurate path-averages. In assessing the performance of the link-derived estimates of rainfall it is therefore important to note that the rain gauge estimates with which they are being compared are themselves approximations.

5.5. Path-Averaging of Radar Data

[44] The available radar data for use in the Bolton study was provided in Cartesian form. To convert the estimated

rain rates into a path-average each Cartesian pixel was weighted in proportion to the length of the link-path passing through it. As with the rain gauge-derived figures, it should be noted that these radar-based estimates of path-averaged rainfall are also only approximations.

6. Event Comparisons

[45] In this section we present three example events. The first (11 February 2001) is an example of a widespread winter stratiform event. The second (17 May 2001) is an event with widespread scattered showers. The third (12–13 September 2001) is a lengthy stratiform event for which there is information from the Hameldon Hill radar.

[46] We consider both the estimated rain rates through the events and the corresponding rainfall accumulations. An idea of the homogeneity (or otherwise) of the rainstorm is provided by the variations in rainfall from gauge to gauge.

[47] The link estimates were obtained using the method described in section 5.3.3, with the reports from the rain gauges nearest to the ends of the link used in place of rain detectors.

6.1. Event 1: 11 February 2001

[48] It rained more or less continuously throughout this day. After some heavy morning showers (with a path-averaged peak of about 6 mm/hr at around 02:00) there were about five hours of light drizzle from 10:00, followed by steady rain (at a rate of about 1.5 mm/hr) for the rest of the day.

[49] In Figure 8 we can see the raw link data plots for this day (single frequencies and their difference). At the peak of the morning storm just under 12 dB of attenuation were recorded at the higher frequency, with a corresponding attenuation difference of about 5 dB.

[50] The four gauges nearest to the link path were all operating reliably and their cumulative plots are shown in Figure 9. All four gauges recorded nearly continuous rain, though the total amounts varied between 28 mm and 43 mm. The differences are mostly in the morning, during which the most northerly gauge recorded over 20 mm of rain compared to the 11 mm recorded by the most southerly gauge.

[51] In Figure 10 (top) we plot the individual 1-minute rain rates derived from the attenuation differences against the corresponding weighted gauge estimates. The line of equality is shown for reference. The agreement at all rain rates is reassuring. At the lowest rain rates the gauge estimates may be expected to be unreliable, since the rain rates are being inferred from tips that are widely separated in time (so that the minute(s) during which the rain in that tip actually fell cannot be accurately identified). The close agreement is a consequence of the nature of the rain falling, which is largely homogeneous in both space and time.

[52] The middle plot of Figure 10 is a time series of the two estimates of path-averaged rain rate. Close agreement is apparent for most of the event. It is easiest to see whether any apparent differences are important by reference to the bottom graph which shows the time series of estimated cumulative rainfall during the event. The two estimates are in close agreement, with the final path-averaged accumulations being 35 mm (gauges) and 31 mm (attenuation

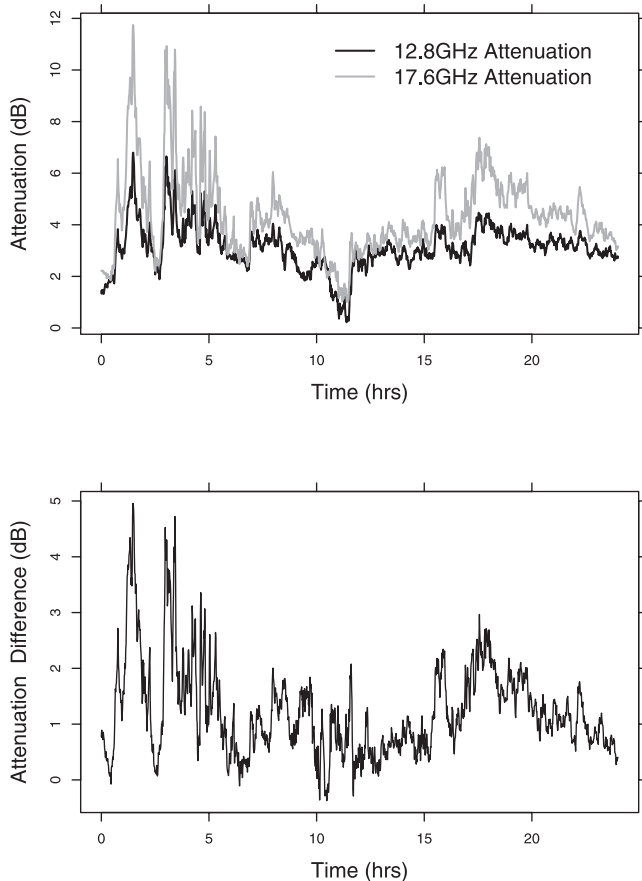


Figure 8. Microwave link attenuations for 11 February 2001: Single frequency attenuations (top) and their difference (bottom).

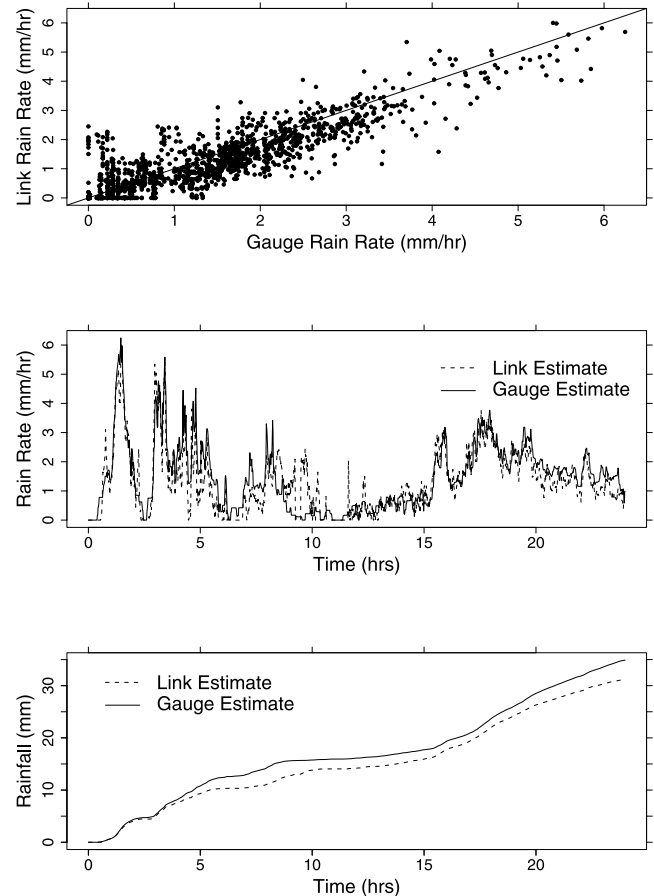


Figure 10. Link and rain gauge estimates of rainfall for the event on 11 February 2001: Link-derived rates against gauge-derived rates (top); time series of link- and gauge-derived rates (middle); time series of link- and gauge-derived accumulations (bottom).

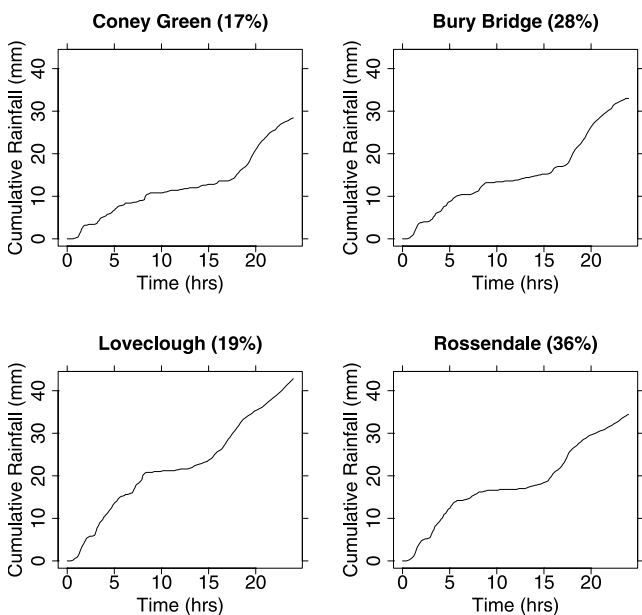


Figure 9. Individual rain gauge accumulations for the event on 11 February 2001: Weights applied to each gauge are given in brackets (%).

difference). It is apparent that most of the discrepancy occurs between around 04:00 and 08:00. This may be due to the Thiessen-based gauge average being unrepresentative in that period (we observed earlier from Figure 9 that there were significant differences in rainfall at the individual gauges during the morning).

6.2. Event 2: 17 May 2001

[53] This six-hour event consisted of several scattered downpours, most of which lasted less than 30 min. The individual gauge time series plots (Figure 11) show huge variations due to localized showers: Rossendale experiences almost 90 mm/hr just before 15:00; Coney Green experiences over 60 mm/hr at around 13:00. Neither heavy burst is recorded at the other gauges, and the result is widely different accumulations: Loveclough records less than 5 mm of rain in total, whereas the other three gauges record totals between 13 mm and 20 mm.

[54] We have already seen the link traces for the day (Figure 3) which showed widely varying attenuations during the period of these storms.

[55] The Thiessen assumption is that rainfall is uniform throughout the polygon surrounding a rain gauge. When there are scattered showers, this assumption is obviously

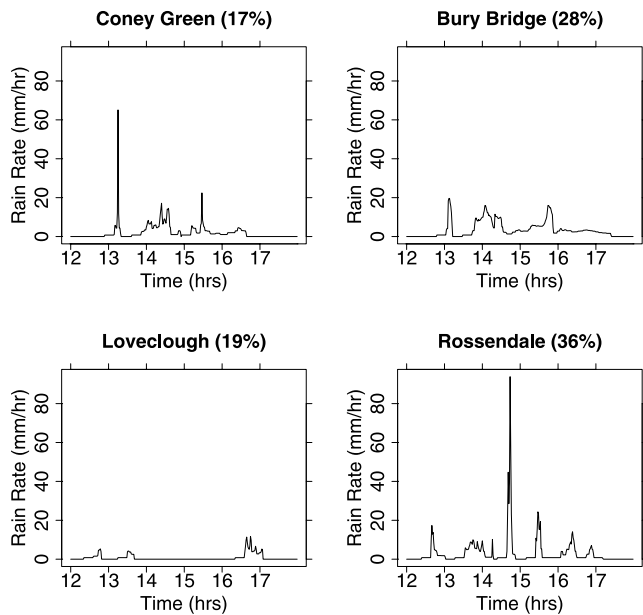


Figure 11. Individual rain gauge traces for the event on 17 May 2001: Weights applied to each gauge are given in brackets (%).

invalid so that the rain gauge-based estimate of the rain falling on the link path is unlikely to be accurate. We can see the effect in Figure 12 (top) in which, although the vast majority of points are in good agreement, there are seven points (i.e., seven individual minutes) for which the weighted-gauge rain rate is far in excess of the corresponding link path-average. These minutes correspond to those in which only one gauge recorded extreme rain rates.

[56] While there may be a few apparent discrepancies between the two sets of estimates of path-averaged rainfall, their importance is slight, as the lower plots in Figure 12 make clear. The estimates of the cumulative rainfall are in close agreement: 15 mm for the gauges and 16 mm for the link.

6.3. Event 3: 12–13 September 2001

[57] The third example considered is a widespread rain event consisting of two fronts, that accumulates nearly 30 mm of rain (according to the gauges) between 09:00 on 12 September and 19:00 on 13 September 2001. A peak path-averaged rain rate of 14 mm/hr was recorded at around 16:00 on 12 September.

[58] Figure 13 shows cumulative estimates obtained using the three methods (link, radar, rain gauge). Agreement is close until just before 16:00 on 12 September. For the next 45 min the radar estimates fall behind the link and gauges which remain in close agreement. Around this time individual gauges record maximum rain rates of between 20 mm/hr and 80 mm/hr, suggesting significant spatial variability during this period. Such small-scale variability might not be visible in radar data that report averages of 2 km × 2 km squares. Better information might have been obtained from the original polar radar data, but these data were not available.

[59] For the remainder of the event it is the radar and link estimates that are in closest agreement. The gauge-derived

rainfall accumulation from the second front (from about 05:00 on 13 September) is only about 14 mm as estimated by the gauges, compared to about 20 mm for the radar and link methods. This may be attributable to the spatial variability of the rainstorm, with the gauges failing to accurately represent the rainfall along the 23.3 km of the link.

7. Summary of Results

[60] The examples presented in section 6 are not special cases: it is generally true that the dual-frequency method results in rainfall estimates that are similar to those obtained from the gauge network.

[61] Figure 14 provides a summary of the level of agreement in accumulations between the attenuation-difference and rain gauge estimates for 112 events between February 2000 and October 2001, amounting to over 1000 hours of rain. The graph reports the results for all events for which the minimum temperature was greater than 5°C. This criterion is designed to exclude events in which sleet occurred (the dual-frequency method does not work in such conditions and rain gauges may not provide reliable records).

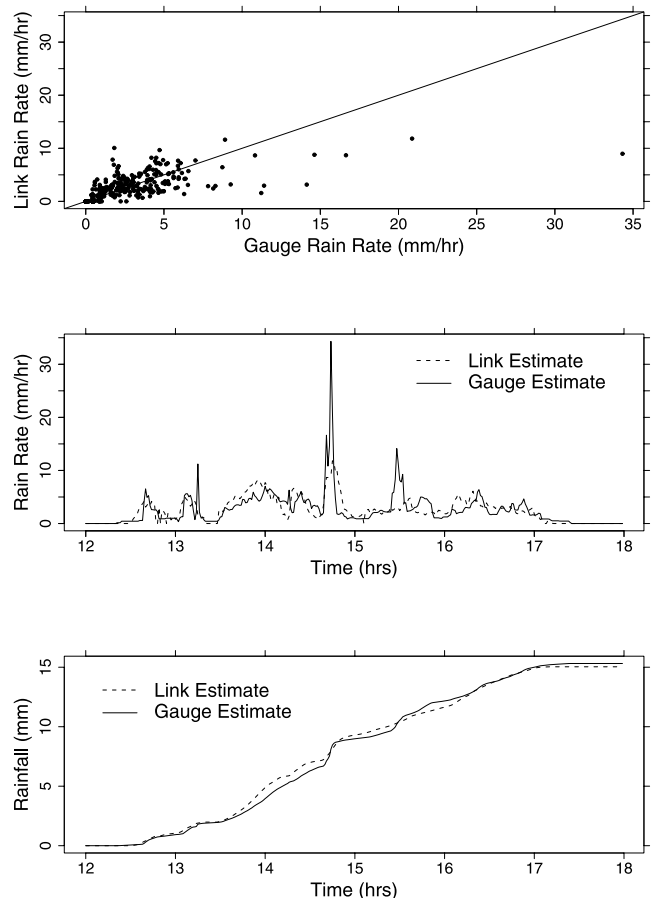


Figure 12. Link and rain gauge estimates of rainfall for the event on 17 May 2001: Link-derived rates against gauge-derived rates (top); time series of link- and gauge-derived rates (middle); time series of link- and gauge-derived accumulations (bottom).

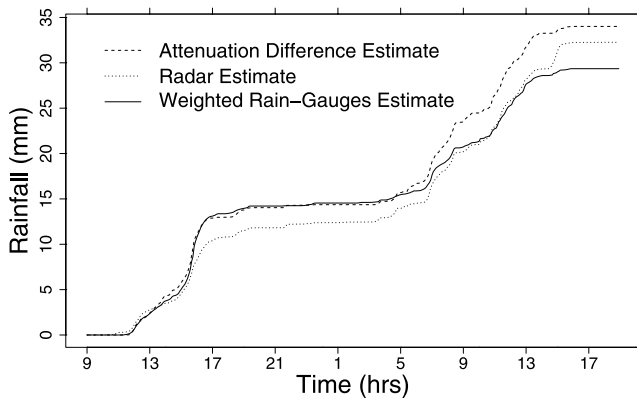


Figure 13. Link, rain gauge and radar estimates of cumulative rainfall for event on 12–13 September 2001.

[62] The majority of estimates (about 60%) lie within the 15% “agreement limits” (i.e., where the link estimate lies within 85% and 115% of the rain gauge estimate). Of the points lying outside these bounds, we are not concerned by those represented as dots since these are cases for which the rainfall recorded by at least one of the individual gauges is more extreme (smaller or greater) than the dual-frequency estimate, indicating spatial rainfall heterogeneity. In such cases it is likely that the gauge-derived average is not properly representative of the rainfall along the link path. This leaves less than one-third of events for which the two estimates differ by more than 15%.

[63] As mentioned previously, only a small amount of radar data is available for comparisons with the link estimates. A summary plot such as Figure 14 is therefore not feasible for the radar-derived estimates. Instead, Table 1 reports link, gauge and radar estimates of six-hour rainfall accumulation for the days from which all three types of data were available. We have excluded periods during which less than 1 mm of rain was recorded. It can be seen that there is no consistent pattern of underestimation or overestimation for any of the three measuring instruments, relative to

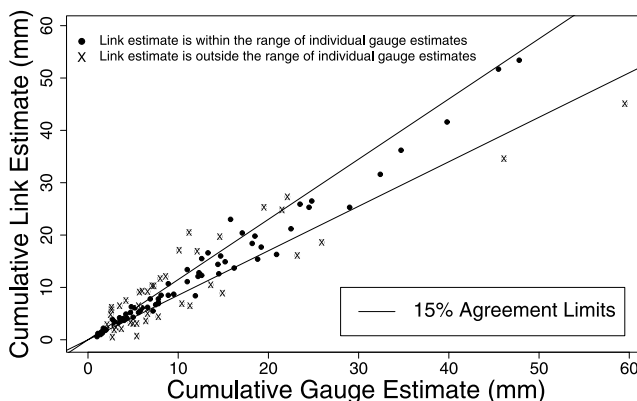


Figure 14. Summary of link and rain gauge accumulation estimates (Hameldon Hill–Clarke’s Hill link; 12.8 GHz and 17.6 GHz): Black dots outside the 15% agreement limits indicate that the rainfall recorded by at least one of the individual gauges was more extreme than the link estimate. Crosses indicate otherwise.

another. This is reassuring since it implies that if we wish to measure rainfall in a situation where rain gauges cannot be employed and radars cannot reach, then, given a clear line-of-sight, a microwave link should be a reliable substitute.

8. Applicability of Link-Derived Estimates

8.1. Link Length

[64] We have seen that by applying the baseline calibration algorithm outlined in section 5, path-averaged rainfall estimates comparable to those from rain gauges and radar can be obtained from a dual-frequency microwave link. However, this conclusion is based on observations from one link of a particular length (23.3 km). A relevant question to be asked is whether the method would work with a link of a different length.

[65] There are no theoretical limitations on link length, only practical considerations that may affect an operational setup. The principal limitation is the need for a clear line-of-sight.

[66] So far as shorter links are concerned, we can report that the rainfall estimates obtained from the 14 km Bolton path from Heather Hall to Clarke’s Hill (see Figure 2), operating at 13.9 GHz and 22.9 GHz, have yielded even closer agreement with the corresponding rain gauge estimates. Using the algorithm for baseline calibration described in section 5, we obtain the link-gauge agreement summary plot in Figure 15. Of the 52 events observed on this link between June 2001 and May 2002, 80% lie within either the 15% agreement limits or the gauge extremes. This link crosses a relatively flat, urban region not subject to orographic effects, so that the reports from the neighboring rain gauges may be more representative of the path of the link.

8.2. Coverage Provided by a Link

[67] The dual-frequency method provides path-averaged estimates of rainfall close to the surface. This is in contrast to rain gauges which provide point estimates at the surface, and radars which provide areal estimates derived from volumes relatively remote from the surface. In any application that involves using rainfall estimates it is necessary to define the region which the estimates are believed to cover.

[68] An idea of the coverage would be given by observing the similarity between the cumulative rainfall measured along a link of length l km with that measured along a

Table 1. Link-Radar-Gauge Comparisons

Start		Estimated 6-hour Accumulations, mm				
Day	Time	Gauge (G)	Link (L)	Radar (R)	100L/G	100R/G
5 Sept	13:00	3.5	2.2	5.2	63	149
5 Sept	19:00	2.6	1.6	2.9	63	111
12 Sept	09:00	4.8	5.5	4.4	114	90
12 Sept	15:00	9.4	8.5	7.4	91	79
13 Sept	03:00	6.1	9.1	7.7	150	127
13 Sept	09:00	8.2	10.0	10.6	121	130
14 Sept	09:00	4.7	6.9	2.3	145	49
14 Sept	15:00	2.6	3.0	2.1	115	79
14 Sept	21:00	3.9	4.2	5.5	106	138
15 Sept	03:00	4.5	2.7	5.6	59	125
15 Sept	09:00	2.4	2.6	4.6	106	189
8 Oct	00:00	20.4	23.3	23.6	114	115
8 Oct	17:00	9.4	10.8	10.5	115	112

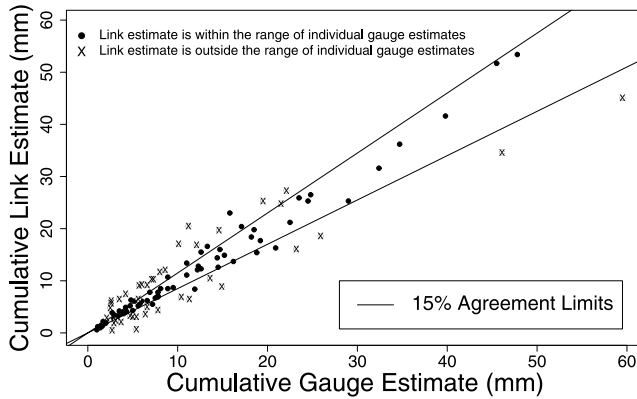


Figure 15. Summary of link and rain gauge accumulation estimates (Heather Hall-Clarke’s Hill link; 13.9 GHz and 22.9 GHz): Black dots outside the 15% agreement limits indicate that the rainfall recorded by at least one of the individual gauges was more extreme than the link estimate. Crosses indicate otherwise.

parallel link, also of length l km, at a distance d km from the original. We can anticipate that the similarity will decrease with increasing d and with decreasing l .

[69] Although we do not have mobile, parallel links with which to investigate this issue, we do have a limited amount of radar data. The results in Table 1 suggest that the Cartesian radar data provide a useful surrogate for link data. We can therefore exploit the radar data as follows. We randomly select parallel 2 km-wide rectangles of length l km from the radar data, with these rectangles referring to hypothetical links at distance d km apart, as illustrated in

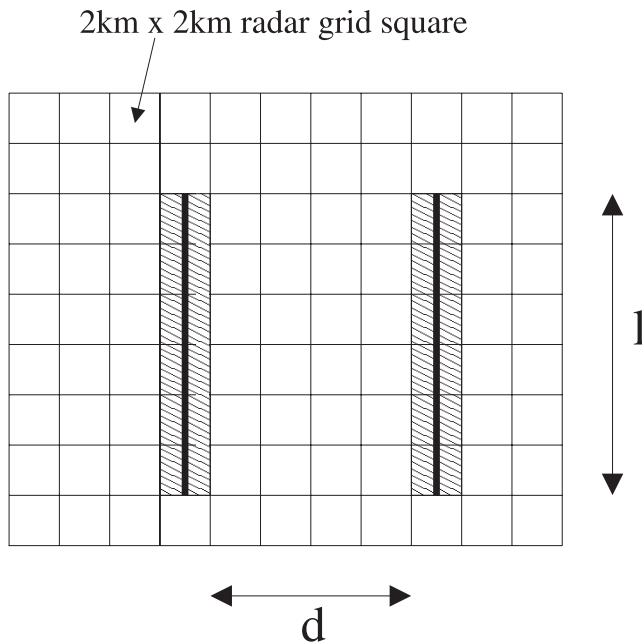


Figure 16. Link coverage diagram: The shaded rectangles made up of adjacent radar squares refer to two hypothetical links. “Path-averaged” rainfall is estimated from the radar data for each hypothetical link. The agreement of the two cumulative estimates is then examined as “link” length (l), distance apart (d) and integration time are varied.

Table 2. Dependence of Link Coverage on Link Length and Integration Time^a

	For APD < 5%	For APD < 15%	For APD < 25%
5-minute totals, 10 km links	$d < 0.05$ km	$d < 0.1$ km	$d < 0.5$ km
5-minute totals, 20 km links	$d < 0.1$ km	$d < 0.25$ km	$d < 0.75$ km
1-hour totals, 10 km links	$d < 0.25$ km	$d < 2$ km	$d < 4$ km
1-hour totals, 20 km links	$d < 0.5$ km	$d < 3$ km	$d < 6$ km
5-hour totals, 10 km links	$d < 0.5$ km	$d < 4$ km	$d < 9$ km
5-hour totals, 20 km links	$d < 1$ km	$d < 6$ km	$d < 12$ km

^aAPD is the approximate percentage difference between the two “link” estimates.

Figure 16. Aggregating the radar data over a period of t minutes for each rectangle (link), we determine the estimated “path-averaged” cumulative rainfall on each of the two hypothetical links. As a measure of agreement between the two values we follow *Kitchen and Blackall* [1992], who compared simultaneous estimates of one-hour rainfall using rain gauges and radar. Writing G for their gauge estimates and R for their radar estimates, their comparisons were based on the value of ϕ , the Root Mean Square Log Ratio given by

$$\phi = \sqrt{\frac{1}{N} \sum_{i=1}^N \{\log_e (R/G)\}^2},$$

with N being the number of events being compared. This statistic can be interpreted by describing

$$100(\exp(\phi) - 1)$$

as the “average percentage difference (APD)” between the two estimates.

[70] For our investigation a thousand pairs of rectangles were chosen at random locations and points in time throughout the 134 hours of available data. In Table 2 we present very approximate results based on the ϕ statistic, with the R and G in the *Kitchen and Blackall* [1992] formula replaced by our two “link” estimates.

[71] As expected the agreement improves with increasing l and t , and with decreasing d . All estimates less than 2 km (the radar grid size) should be regarded as particularly speculative (being based on interpolation of a regression line fitted to values of $d = 0, 2, 4, \dots$). (1) For a 1-hour storm falling on the Bolton sub-catchment (21 km²), a centrally placed 10 km link should be able to achieve an APD of less than 15%. (2) For a 5-hour storm falling on a catchment of 10 km by 10 km, a centrally placed link ought to be able to achieve an APD of around 15%. (3) For a 5-hour storm falling on a 20 km link, the APD of 25% might provide useful information for as much as 500 km². The accuracy quoted is derived from sampling three storms in the area covered by the Hameldon Hill radar. Greater accuracy might be expected in flatter regions, and less in more hilly areas, because of differences in orography.

9. Conclusions and Further Work

[72] The precise form of the relation between rainfall and the attenuation of microwaves has long been a subject of

concern to those wishing to infer the one from the other. A strength of the method proposed here is that it is not necessary to make assumptions about the drop size distribution (or other rainfall characteristics).

[73] To implement the method it is necessary to correctly determine the baseline attenuation levels. This was achieved by reference to the mean values taken from the nearest dry periods. These periods were identified by study of the correlation between the two frequencies (and, ideally, by the use of rain detectors). The method does not require information on other variables.

[74] The results presented in this paper have shown that reliable estimates of path-averaged rainfall can be obtained using the difference in the attenuation experienced by two microwave links using well-chosen frequencies. In terms of comparability with gauge-derived estimates of rainfall, the estimates obtained are generally within 15% of those obtained using rain gauges. Reasonable agreement has also been observed with radar-derived estimates.

[75] The potential for real-time rainfall estimation using the attenuation difference method is evident. The technique should be of particular use in regions where gauges are difficult to locate, or where radars cannot reach (e.g., because of obscuration).

[76] In order to further investigate the dual-frequency technique, a follow-up experiment entitled MANTISSA (Microwave Attenuation as a New Tool for Improving Storm water Supervision Administration) is currently in progress. As well as continuing data collection from the Bolton links, additional dual-frequency microwave links have been installed in Germany and two locations in Italy. These links vary in length between 3 km and 29 km and operate at different pairs of frequencies. The results of this experiment, when available, should provide further information on the potential use of dual-frequency links to estimate rainfall, including further investigation of the issues of link length and coverage, addressed in section 8. The principal applications of links to be considered in the MANTISSA project are as input data for hydrological models, and for online calibration of radar data. More information regarding the experiment can be found at <http://prswwww.essex.ac.uk/mantissa/>.

[77] **Acknowledgments.** This work has been supported by the UK Natural Environment Research Council through grant GR3/CO035. We gratefully acknowledge the collaboration in the experiment of John Goddard and Mike Willis at the Radiocommunications Research Unit at the Rutherford Appleton Laboratory and of Chris Collier and his colleagues at the Telford Institute of Environmental Systems, University of Salford. We have had industrial support from United Utilities plc, the UK Environment Agency, the UK Met Office, Crown Castle International and Your Communications, to whom grateful thanks are also due.

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